Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-18 EM Wide Berm 2% - Test 1 - .experimental data (4) .txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.250 Hz*
Freq. upper bound:	*5.000 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*33*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*38*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *41.5%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1243*	*0.05164*
Peak wave period T_P (s):	*2*	*2*
Energy wave period $T_{-1,0}$ (s):	*1.778*	*1.897*
Mean wave period $T_{0,1}$ (s):	*1.598*	*1.544*
Mean wave period $T_{0,2}$ (s):	*1.483*	*1.301*
Spectral width (Broadness):	*0.7461*	*0.8689*
Spectral width (Narrowness):	*0.4001*	*0.6395*

Time Domain Analysis

Overall amplitude reflection coefficient: *41.7%*

	Incident	**Reflected**
Number of waves:	1015	982
Mean wave height H_m (m):	*0.07955*	*0.03425*
Mean wave period T_m (s):	*1.674*	*1.73*
Sig. wave height H_s (m):	*0.1234*	*0.05389*
$T_{H_{1/3}}$ (s):	*1.902*	*2.022*
H_{max} (m):	*0.21*	*0.1217*
T_{Hmax} (s):	*2.165*	*1.89*
$H_{1/10}$ (m):	*0.1565*	*0.06737*
$H_{1/50}$ (m):	*0.1866*	*0.08204*
$H_{1/100}$ (m)	*0.195*	*0.08781*
$H_{1/250}$ (m):	*0.2049*	*0.09828*
$H_{0.1\%}$ (m):	*0.2099*	*Not enough data*
$H_{1\%}$ (m):	*0.1847*	*0.07892*
$H_{2\%}$ (m):	*0.1713*	*0.0736*
$H_{10\%}$ (m)	*0.134*	*0.05727*
Groupiness factor GF:	*1.095*	*1.086*
Skewness b1:	*0.7213*	*0.08523*
Kurtosis b2:	*3.459*	*3.049*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-18 EM Wide Berm 2% - Test 1 - .experimental (4) .txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap (% of blocks):	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *42.8%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1368*	*0.05858*
Peak wave period T_P (s):	*2.065*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.815*	*1.961*
Mean wave period $T_{0,1}$ (s):	*1.595*	*1.52*
Mean wave period $T_{0,2}$ (s):	*1.459*	*1.24*
Spectral width (Broadness):	*0.7825*	*0.8815*
Spectral width (Narrowness):	*0.4412*	*0.7098*

Time Domain Analysis

Overall amplitude reflection coefficient: *42.8%*

	Incident	Reflected
Number of waves:	1027	1016
Mean wave height H_m (m):	*0.08992*	*0.03947*
Mean wave period T_m (s):	*1.742*	*1.76*
Sig. wave height H_s (m):	*0.1391*	*0.06079*
$T_{H_{1/3}}$ (s):	*1.97*	*2.05*
H_{max} (m):	*0.2261*	*0.1008*
T_{Hmax} (s):	*1.584*	*1.183*
$H_{1/10}$ (m):	*0.17*	*0.0737*
$H_{1/50}$ (m):	*0.1921*	*0.08503*
$H_{1/100}$ (m)	*0.1982*	*0.08917*
$H_{1/250}$ (m):	*0.2063*	*0.09343*
$H_{0.1\%}$ (m):	*0.2254*	*0.1007*
$H_{1\%}$ (m):	*0.1908*	*0.08328*
$H_{2\%}$ (m):	*0.1829*	*0.07893*
$H_{10\%}$ (m)	*0.1493*	*0.06582*
Groupiness factor GF:	*1.083*	*1.004*
Skewness b1:	*0.8159*	*0.03815*
Kurtosis b2:	*3.467*	*2.794*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-18 EM Wide Berm 2% - Test 1 - .experimental data.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.227 Hz*
Freq. upper bound:	*4.531 Hz*

Freq. Domain Analysis

Overlap 20% (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*35*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*42*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *44.6%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1463*	*0.06521*
Peak wave period T_P (s):	*2.207*	*2.207*
Energy wave period $T_{-1,0}$ (s):	*1.888*	*2.068*
Mean wave period $T_{0,1}$ (s):	*1.632*	*1.545*
Mean wave period $T_{0,2}$ (s):	*1.476*	*1.232*
Spectral width (Broadness):	*0.8058*	*0.8875*
Spectral width (Narrowness):	*0.4707*	*0.757*

Time Domain Analysis

Overall amplitude reflection coefficient: *44.6%*

	Incident	**Reflected**
Number of waves:	1018	1037
Mean wave height H_m (m):	*0.09712*	*0.0436*
Mean wave period T_m (s):	*1.813*	*1.782*
Sig. wave height H_s (m):	*0.1492*	*0.0661*
$T_{H_{1/3}}$ (s):	*2.074*	*2.121*
H_{max} (m):	*0.2482*	*0.1165*
T_{Hmax} (s):	*1.825*	*2.127*
$H_{1/10}$ (m):	*0.181*	*0.07939*
$H_{1/50}$ (m):	*0.2005*	*0.09156*
$H_{1/100}$ (m)	*0.2081*	*0.0961*
$H_{1/250}$ (m):	*0.2195*	*0.1037*
$H_{0.1\%}$ (m):	*0.2476*	*0.116*
$H_{1\%}$ (m):	*0.1986*	*0.08906*
$H_{2\%}$ (m):	*0.19*	*0.08604*
$H_{10\%}$ (m)	*0.1634*	*0.06988*
Groupiness factor GF:	*1.061*	*0.9641*
Skewness b1:	*0.8402*	*-0.01657*
Kurtosis b2:	*3.413*	*2.703*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-18 EM Wide Berm 2% - Test 1 - .experimental data*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.211 Hz*
Freq. upper bound:	*4.219 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*46*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *37.5%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.162*	*0.06076*
Peak wave period T_P (s):	*2.286*	*2.37*
Energy wave period $T_{-1,0}$ (s):	*1.955*	*2.02*
Mean wave period $T_{0,1}$ (s):	*1.628*	*1.248*
Mean wave period $T_{0,2}$ (s):	*1.441*	*0.9711*
Spectral width (Broadness):	*0.8339*	*0.8503*
Spectral width (Narrowness):	*0.5261*	*0.8071*

Time Domain Analysis

Overall amplitude reflection coefficient: *37.5%*

	Incident	Reflected
Number of waves:	1033	1276
Mean wave height H_m (m):	*0.1127*	*0.04055*
Mean wave period T_m (s):	*1.904*	*1.541*
Sig. wave height H_s (m):	*0.1668*	*0.06213*
$T_{H_{1/3}}$ (s):	*2.187*	*1.919*
H_{max} (m):	*0.2472*	*0.1057*
T_{Hmax} (s):	*1.605*	*2.718*
$H_{1/10}$ (m):	*0.1941*	*0.07558*
$H_{1/50}$ (m):	*0.2193*	*0.08873*
$H_{1/100}$ (m)	*0.2277*	*0.09361*
$H_{1/250}$ (m):	*0.2365*	*0.09792*
$H_{0.1\%}$ (m):	*0.2471*	*0.1045*
$H_{1\%}$ (m):	*0.2191*	*0.08909*
$H_{2\%}$ (m):	*0.2063*	*0.08171*
$H_{10\%}$ (m)	*0.1757*	*0.06634*
Groupiness factor GF:	*1.026*	*1.032*
Skewness b1:	*0.9324*	*-0.1594*
Kurtosis b2:	*3.368*	*3.035*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-18 EM Wide Berm 2% - Test 1 - .experimental.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.289 Hz*
Freq. upper bound:	*5.781 Hz*

Freq. Domain Analysis

Overlap of (4) series:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*30*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*33*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *39.2%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1083*	*0.04247*
Peak wave period T_P (s):	*1.73*	*1.882*
Energy wave period $T_{-1,0}$ (s):	*1.618*	*1.684*
Mean wave period $T_{0,1}$ (s):	*1.495*	*1.426*
Mean wave period $T_{0,2}$ (s):	*1.416*	*1.245*
Spectral width (Broadness):	*0.6816*	*0.8383*
Spectral width (Narrowness):	*0.3398*	*0.5589*

Time Domain Analysis

Overall amplitude reflection coefficient: *39.1%*

	Incident	**Reflected**
Number of waves:	1031	1041
Mean wave height H_m (m):	*0.06794*	*0.02728*
Mean wave period T_m (s):	*1.52*	*1.505*
Sig. wave height H_s (m):	*0.1059*	*0.04407*
$T_{H_{1/3}}$ (s):	*1.687*	*1.786*
H_{max} (m):	*0.1892*	*0.07803*
T_{Hmax} (s):	*1.687*	*1.886*
$H_{1/10}$ (m):	*0.1319*	*0.05687*
$H_{1/50}$ (m):	*0.1551*	*0.069*
$H_{1/100}$ (m)	*0.1631*	*0.07249*
$H_{1/250}$ (m):	*0.1719*	*0.07616*
$H_{0.1\%}$ (m):	*0.1885*	*0.078*
$H_{1\%}$ (m):	*0.1536*	*0.06826*
$H_{2\%}$ (m):	*0.1434*	*0.06453*
$H_{10\%}$ (m)	*0.1126*	*0.04789*
Groupiness factor GF:	*1.058*	*1.148*
Skewness b1:	*0.577*	*0.1935*
Kurtosis b2:	*3.225*	*3.36*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-1Emerged Wide Berm 2% - Test 1 - .experiment output (4)*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of blocks (%):	*50% (4)*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*27*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *33.7%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.08229*	*0.02774*
Peak wave period T_P (s):	*1.524*	*1.524*
Energy wave period $T_{-1,0}$ (s):	*1.441*	*1.449*
Mean wave period $T_{0,1}$ (s):	*1.364*	*1.275*
Mean wave period $T_{0,2}$ (s):	*1.315*	*1.154*
Spectral width (Broadness):	*0.5935*	*0.7907*
Spectral width (Narrowness):	*0.2771*	*0.4697*

Time Domain Analysis

Overall amplitude reflection coefficient: *33.7%*

	Incident	**Reflected**
Number of waves:	1032	1094
Mean wave height H_m (m):	*0.05233*	*0.01758*
Mean wave period T_m (s):	*1.362*	*1.286*
Sig. wave height H_s (m):	*0.08031*	*0.02759*
$T_{H_{1/3}}$ (s):	*1.482*	*1.53*
H_{max} (m):	*0.129*	*0.04813*
T_{Hmax} (s):	*1.517*	*1.732*
$H_{1/10}$ (m):	*0.09698*	*0.03468*
$H_{1/50}$ (m):	*0.1125*	*0.04174*
$H_{1/100}$ (m)	*0.1164*	*0.04422*
$H_{1/250}$ (m):	*0.1213*	*0.04645*
$H_{0.1\%}$ (m):	*0.1289*	*0.04803*
$H_{1\%}$ (m):	*0.1111*	*0.0417*
$H_{2\%}$ (m):	*0.1067*	*0.03786*
$H_{10\%}$ (m)	*0.08523*	*0.02946*
Groupiness factor GF:	*0.957*	*1.005*
Skewness b1:	*0.3313*	*0.3069*
Kurtosis b2:	*2.899*	*3.026*

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Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-Emerged Wide Berm 2% - Test 1 - .experimental - 1300 [s] (4) [ex]*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

Freq. Domain Analysis

Overlap:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*37*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *45%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1574*	*0.07079*
Peak wave period T_P (s):	*2.286*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.915*	*2.111*
Mean wave period $T_{0,1}$ (s):	*1.623*	*1.508*
Mean wave period $T_{0,2}$ (s):	*1.452*	*1.178*
Spectral width (Broadness):	*0.8212*	*0.8878*
Spectral width (Narrowness):	*0.4998*	*0.8002*

Time Domain Analysis

Overall amplitude reflection coefficient: *45%*

	Incident	**Reflected**
Number of waves:	1030	1043
Mean wave height H_m (m):	*0.1066*	*0.04897*
Mean wave period T_m (s):	*1.847*	*1.824*
Sig. wave height H_s (m):	*0.1616*	*0.07118*
$T_{H_{1/3}}$ (s):	*2.098*	*2.133*
H_{max} (m):	*0.232*	*0.109*
T_{Hmax} (s):	*1.712*	*2.063*
$H_{1/10}$ (m):	*0.1905*	*0.0845*
$H_{1/50}$ (m):	*0.2124*	*0.09538*
$H_{1/100}$ (m)	*0.2205*	*0.09863*
$H_{1/250}$ (m):	*0.2287*	*0.1035*
$H_{0.1\%}$ (m):	*0.232*	*0.1088*
$H_{1\%}$ (m):	*0.2101*	*0.09419*
$H_{2\%}$ (m):	*0.2003*	*0.08995*
$H_{10\%}$ (m)	*0.1742*	*0.07553*
Groupiness factor GF:	*1.061*	*0.9057*
Skewness b1:	*0.923*	*-0.06966*
Kurtosis b2:	*3.471*	*2.615*

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